Quadratic Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 9$ in solving multi-step equations, evaluate $f(5)$.

Final Answer: 34. Substitute $x = 5$: $f(5) = 5(5) + 9 = 34$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 3$ in literal equations and formulas, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for literal equations and formulas consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 4$ in practical problems with equations, evaluate $f(2)$.

Final Answer: 8. Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for practical problems with equations consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 3x + 5$ in review: solving one-step inequalities, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 3(2) + 5 = 11$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for review: solving one-step inequalities consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

Identify coefficients: $a = 2$, $b = -6$, $c = 5$. Evaluate the discriminant: $\Delta = b^2 - 4ac = (-6)^2 - 4(2)(5) = 36 - 40 = -4$. Because \(\Delta

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Factor the quadratic expression $6x^2 + 11x - 10$ completely.

Use the ac-method. We seek two integers whose product is $a \cdot c = 6(-10) = -60$ and whose sum is $b = 11$. The factor pair of -60 that sums to 11 is 15 and -4 (since $15 \times (-4) = -60$ and $15 + (-4) = 11$). Rewrite the middle term and factor by grouping: $6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5)$. Factor out the binomial $(2x + 5)$: $(2x + 5)(3x - 2)$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given $f(x) = \frac{2}{x-1}$ and $g(x) = \sqrt{x+5}$, find (a) the value of $(f \circ g)(4)$, and (b) the domain of $g(x)$.

1. Evaluate the inner function first: $g(4) = \sqrt{4+5} = \sqrt{9} = 3$. 2. Substitute into f: $(f \circ g)(4) = f(g(4)) = f(3) = \frac{2}{3-1} = \frac{2}{2} = 1$. 3. Domain of $g(x) = \sqrt{x+5}$: The radicand must be non-negative: $x + 5 \geq 0 \Rightarrow x \geq -5$ or in interval notation $[-5, \infty)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after t hours, and find the population after 12 hours.

Initial population $a = 250$, doubling factor $b = 2$, time interval = 3 hours: $B(t) = 250 \cdot 2^{t/3}$. For $t = 12$ hours, the number of doubling periods is $12/3 = 4$: $B(12) = 250 \cdot 2^4 = 250 \cdot 16 = 4000$ bacteria.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions & Transformations · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

Apply power of a power to numerator and denominator separately: Numerator $= 2^3 \cdot (x^3)^3 \cdot (y^{-2})^3 = 8x^9y^{-6}$. Denominator $= 4^2 \cdot (x^{-1})^2 \cdot (y^4)^2 = 16x^{-2}y^8$. Divide coefficients and subtract exponents: $\frac{8}{16} \cdot x^{9-(-2)} \cdot y^{-6-8} = \frac{1}{2}x^{11}y^{-14} = \frac{x^{11}}{2y^{14}}$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Factoring Polynomials

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5}-\sqrt{2}}$.

Multiply numerator and denominator by the conjugate $\sqrt{5}+\sqrt{2}$: $\frac{\sqrt{3}(\sqrt{5}+\sqrt{2})}{(\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2})}$. Expand denominator using difference of squares $(a-b)(a+b) = a^2 - b^2$: $(\sqrt{5})^2 - (\sqrt{2})^2 = 5 - 2 = 3$. Expand numerator and simplify: $\frac{\sqrt{15}+\sqrt{6}}{3}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 9.01 Radicals & Exponents

Factor the polynomial completely by grouping: $2x^3 - 3x^2 - 8x + 12$.

Group terms in pairs: $(2x^3 - 3x^2) - (8x - 12)$. Factor out the greatest common factor from each pair: $x^2(2x - 3) - 4(2x - 3)$. Factor out the common binomial $(2x - 3)$: $(x^2 - 4)(2x - 3)$. Recognize $(x^2 - 4)$ as a difference of squares and factor completely: $(x - 2)(x + 2)(2x - 3)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Linear Equations & Lines

Solve the linear inequality $-3(2x - 4) \le 4x - 8$ and express the solution in interval notation.

Expand the left side: $-6x + 12 \le 4x - 8$. Collect all x terms on one side: $-10x \le -20$. Divide both sides by -10 and reverse the inequality symbol: $x \ge 2$. In interval notation: $[2, \infty)$.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Rational Expressions

[Exponents & Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Statistics & Data Analysis

[Factoring Polynomials · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 7$ in review: solving one- and two- step equations, evaluate $f(2)$.

a. 13 ✓

b. 16

c. 6

d. 17

Substitute $x = 2$: $f(2) = 3(2) + 7 = 13$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in properties of equality, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: solving one- and two- step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for properties of equality consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The graph of the quadratic function $f(x) = ax^2 + bx + c$ passes through the points $(1, 5)$, $(-1, 1)$, and $(2, 10)$. Determine the values of a, b, and c.

a. $a = 2, b = -1, c = 4$

b. $a = 1, b = 2, c = 2$

c. $a = 3, b = -2, c = 4$

d. $a = 2, b = 1, c = 2$ ✓

Let the quadratic function be $f(x) = ax^2 + bx + c$. We are given three points that the graph passes through: 1. For point $(1, 5)$: $f(1) = a(1)^2 + b(1) + c = a + b + c = 5$ (Equation 1) 2. For point $(-1, 1)$: $f(-1) = a(-1)^2 + b(-1) + c = a - b + c = 1$ (Equation 2) 3. For point $(2, 10)$: $f(2) = a(2)^2 + b(2) + c = 4a + 2b + c = 10$ (Equation 3) Subtract Equation 2 from Equation 1: $(a + b + c) - (a - b + c) = 5 - 1$ $2b = 4$ $b = 2$ Substitute $b = 2$ into Equation 1 and Equation 2: Equation 1 becomes.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following equations represents a parabola that opens downwards?

a. $y = 2x^2 + 3x - 1$

b. $y = -x^2 + 5$ ✓

c. $y = x^2 - 4x$

d. $y = 3x^2$

A parabola opens downwards if the coefficient of the x^2 term is negative. In the given options: - Option A: The coefficient of x^2 is 2 (positive), so it opens upwards. - Option B: The coefficient of x^2 is -1 (negative), so it opens downwards. - Option C: The coefficient of x^2 is 1 (positive), so it opens upwards. - Option D: The coefficient of x^2 is 3 (positive), so it opens upwards. Therefore, $y = -x^2 + 5$ represents a parabola that opens downwards.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic function $f(x) = 2x^2 - 8x + 6$. What is the y-intercept of the graph of this function?

a. -8

b. **6** ✓

c. 2

d. -4

The y-intercept of a function $f(x)$ is the value of the function when $x = 0$. For the quadratic function $f(x) = 2x^2 - 8x + 6$, we substitute $x = 0$ into the equation: $f(0) = 2(0)^2 - 8(0) + 6$ $f(0) = 0 - 0 + 6$ $f(0) = 6$ Therefore, the y-intercept is 6. This corresponds to the constant term in the quadratic equation when it is in the standard form $ax^2 + bx + c$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the larger root of $x^2 - 2x - 48 = 0$?

a. 9

b. **8** ✓

c. 7

d. -6

Factors as $(x - (8))(x - (-6)) = 0$, so $x = 8$ or $x = -6$; the larger is 8.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the x-coordinate of the vertex of $f(x) = -1 - x^2 + 3x$.

a. $\frac{3}{2}$ ✓

b. $\frac{5}{2}$

c. 3

d. $-\frac{3}{2}$

Axis of symmetry: $x = \frac{-b}{2a} = \frac{-(3)}{2(-1)} = \frac{3}{2}$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Let $f(x) = e^{2x} - 5e^x + 6$. The minimum value of $f(x)$ occurs at a point $x = a$. If the equation $f(x) = k$ has exactly one real solution for x, find the value of k.

a. $k = 0$

b. $k = -1/4$ ✓

c. $k = 1/4$

d. $k = 1$

Let $u = e^x$. Then $f(x) = u^2 - 5u + 6$. Since x is real, $u = e^x > 0$. We are looking for the minimum value of the quadratic $g(u) = u^2 - 5u + 6$ for $u > 0$. The vertex of the parabola $g(u)$ occurs at $u = -\frac{-5}{2(1)} = \frac{5}{2}$. Since $\frac{5}{2} > 0$, the minimum value of $g(u)$ for $u > 0$ occurs at $u = \frac{5}{2}$. This corresponds to $e^x = \frac{5}{2}$, so $x = \ln(\frac{5}{2})$. This is the value of a. The minimum value of $f(x)$ is.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the coordinate plane shown, triangle $A'B'C'$ is obtained by first reflecting triangle ABC across the line $y = x$, and then translating the resulting image by h units horizontally and k units vertically. What is the ordered pair (h, k)?

a. $(-4, -2)$ ✓

b. $(-3, -3)$

c. $(-2, -4)$

d. $(4, 2)$

1. Identify the coordinates of original triangle ABC from the grid: $A(1, 2)$, $B(4, 2)$, and $C(2, 5)$. 2. Identify the coordinates of the final image triangle $A'B'C'$: $A'(-2, -1)$, $B'(-2, 2)$, and $C'(1, 0)$. 3. Apply the reflection across the line $y = x$. The rule for reflection across $y = x$ maps each point (x, y) to (y, x). Applying this to ΔABC yields the intermediate vertices A_0, B_0, C_0: - $A(1, 2) \to A_0(2, 1)$ - $B(4, 2) \to B_0(2, 4)$.

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32 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies the following properties: 1. The graph of $y = f(x)$ is symmetric with respect to the line $x = 2$. 2. The graph of $y = f(x)$ is symmetric with respect to the line $x = 5$. 3. For all $x \in [2, 5]$, $f(x) = -(x - 3)^2 + 4$. Let $g(x) = f(x - 1) + 1$. How many real solutions does the equation $g(x) = \frac{1}{3}x$ have for $x > 0$?

a. 3

b. 4

c. 5 ✓

d. 6

We first analyze the symmetry and periodicity of $f(x)$: 1. Reflection across $x = 2$ means $f(x) = f(4 - x)$. 2. Reflection across $x = 5$ means $f(x) = f(10 - x)$. Combining these two reflections yields a translation: $f(x + 6) = f(10 - (4 - x)) = f(4 - x) = f(x)$ Thus, f is periodic with fundamental period $T = 2(5 - 2) = 6$. On the interval $[2, 5]$, $f(x) = -(x - 3)^2 + 4$. Using the symmetry about $x = 2$, for $x \in [-1, 2]$: $f(x) = f(4 - x) = -(4 - x - 3)^2 + 4 = -(x - 1)^2 + 4$ Therefore, over one full.

33 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Which of the following is the completely factored form of the quadratic expression $x^2 - 5x - 14$?

a. $(x - 7)(x + 2)$ ✓

b. $(x + 7)(x - 2)$

c. $(x - 14)(x + 1)$

d. $(x - 7)(x - 2)$

To factor the quadratic expression $x^2 - 5x - 14$ into the form $(x + a)(x + b)$, we must find two integers a and b such that: 1. Their product is the constant term: $a \cdot b = -14$ 2. Their sum is the linear coefficient: $a + b = -5$ We examine pairs of integer factors of -14: - 1 and -14: sum is -13 - -1 and 14: sum is 13 - 2 and -7: sum is $2 + (-7) = -5$ - -2 and 7: sum is 5 The pair that satisfies both conditions is -7 and 2. Therefore, the factored form is $(x - 7)(x + 2)$..

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An exponential function is given by $f(x) = a \cdot b^x$, where a and b are positive constants. The graph of $y = f(x)$ in the xy-plane passes through the points $(0, 4)$ and $(2, 36)$. What is the value of b?

a. 3 ✓

b. 6

c. 9

d. 16

To find the value of b, we use the coordinates of the two given points: 1. Substitute $(0, 4)$ into $f(x) = a \cdot b^x$: $f(0) = a \cdot b^0 = a \cdot 1 = 4 \Rightarrow a = 4$ 2. Substitute $a = 4$ and the point $(2, 36)$ into the function: $f(2) = 4 \cdot b^2 = 36$ 3. Solve for b: $b^2 = \frac{36}{4} = 9$ Since b is given as a positive constant, $b = \sqrt{9} = 3$. Misconception Analysis: - Choice B (6) results from mistakenly setting $b^2 = 36$ without dividing by $a = 4$. - Choice C (9) results from s.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The graph of the function $y = f(x)$ is transformed to produce the graph of $y = g(x)$. The graph of $y = g(x)$ is obtained by applying the following transformations to the graph of $y = f(x)$ in order: a vertical stretch by a factor of 3, followed by a horizontal shift of 4 units to the left, and finally a reflection in the x-axis. Given that $f(x) = x^2$, determine the equation of $y = g(x)$.

a. $g(x) = -(x + 4)^2 + 3$

b. $g(x) = -3(x + 4)^2$ ✓

c. $g(x) = -(x - 4)^2 + 3$

d. $g(x) = -3(x - 4)^2$

Let the original function be $f(x) = x^2$. 1. **Vertical stretch by a factor of 3**: This transformation changes $y = f(x)$ to $y = 3f(x)$. So, the intermediate function is $y = 3x^2$. 2. **Horizontal shift of 4 units to the left**: This transformation changes $y = h(x)$ to $y = h(x + 4)$. Applying this to $y = 3x^2$, we get $y = 3(x + 4)^2$. 3. **Reflection in the x-axis**: This transformation changes $y = k(x)$ to $y = -k(x)$. Applying this to $y = 3(x + 4)^2$, we get $y = -3(x + 4)^2$. Therefore, the equation of.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which of the following represents the exponential function $f(x) = b^x$ where the base b is greater than 1?

a. A graph that is always decreasing and passes through (0,1)

b. **A graph that is always increasing and passes through (0,1)** ✓

c. A graph that is always decreasing and passes through (0,0)

d. A graph that is always increasing and passes through (0,0)

An exponential function of the form $f(x) = b^x$ where the base $b > 1$ is characterized by two key features: it always passes through the point (0,1) because any non-zero number raised to the power of 0 is 1 (i.e., $b^0 = 1$), and it is always increasing. As x increases, $f(x)$ increases at an accelerating rate. Conversely, if $0 < b < 1$, the function would be decreasing. Therefore, the correct representation is a graph that is always increasing and passes through (0,1).

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given the exponential function $f(x) = 5^x$, what is the value of $f(-2)$?

a. $\frac{1}{25}$ ✓

b. -10

c. 25

d. $-\frac{1}{10}$

To find the value of $f(-2)$ for the function $f(x) = 5^x$, we substitute $x = -2$ into the function: $f(-2) = 5^{-2}$. Recall that a negative exponent means taking the reciprocal of the base raised to the positive exponent. Therefore, $5^{-2} = \frac{1}{5^2}$. Calculating 5^2 gives 25. Thus, $f(-2) = \frac{1}{25}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Linear Equations & Lines

The graph of the function $y = f(x)$ is shown below. Which of the following represents the graph of $y = f(x + 2)$?

a. Graph A

b. **Graph B ✓**

c. Graph C

d. Graph D

The transformation $y = f(x + 2)$ represents a horizontal shift of the graph of $y = f(x)$ by 2 units to the left. If a point (x, y) is on the graph of $y = f(x)$, then the point $(x - 2, y)$ is on the graph of $y = f(x + 2)$. Therefore, we need to find the graph that shows the original graph shifted 2 units to the left.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 10.01 Rational Expressions

Which of the following is the correct factored form of the quadratic expression $x^2 - 5x + 6$?

a. $(x - 2)(x - 3)$ ✓

b. $(x + 2)(x + 3)$

c. $(x - 1)(x - 6)$

d. $(x + 1)(x + 6)$

To factor the quadratic expression $x^2 - 5x + 6$, we need to find two numbers that multiply to 6 and add up to -5. These two numbers are -2 and -3. Therefore, the factored form of the expression is $(x - 2)(x - 3)$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 11.01 Statistics & Data Analysis

A function is defined by $f(x) = 2x + 3$. What is the value of $f(4)$?

a. 10

b. **11 ✓**

c. 7

d. 14

To find the value of $f(4)$, substitute $x = 4$ into the function $f(x) = 2x + 3$. $f(4) = 2(4) + 3$ $f(4) = 8 + 3$ $f(4) = 11$

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Exponents & Polynomials

A population starts at 300 and multiplies by 2 every year. What is the population after 2 years?

a. **1200 ✓**

b. 2400

c. 600

d. 1500

$300 \cdot 2^2 = 300 \cdot 4 = 1200$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Functions & Transformations

Which is the inverse of $y = e^x$?

a. $y = \frac{1}{e^x}$

b. $y = \ln x$ ✓

c. $y = \log x$

d. $y = e^{-x}$

The inverse of $y = e^x$ is $y = \ln x$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

The graph of the quadratic function $f(x) = \frac{1}{2}x^2 - 2$ is shown in the figure below. How many distinct real solutions does the equation $(f \circ f \circ f)(x) = 0$ have?

a. 3

b. 4 ✓

c. 6

d. 8

We solve the iterated equation $f(f(f(x))) = 0$ layer by layer from the outside in. **Step 1: Outer Layer** Let $u = (f \circ f)(x)$. The equation becomes: $f(u) = 0 \Rightarrow \frac{1}{2}u^2 - 2 = 0 \Rightarrow u^2 = 4 \Rightarrow u = -2$ or $u = 2$ **Step 2: Middle Layer** Now set $v = f(x)$ and solve $f(v) = u$ for each value of u: - Case A: $f(v) = -2$ $\frac{1}{2}v^2 - 2 = -2 \Rightarrow \frac{1}{2}v^2 = 0 \Rightarrow v = 0$ Since -2 is the minimum value (vertex) of the parabola, this y.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Systems & Inequalities

The polynomial $P(x) = 2x^4 - 7x^3 + ax^2 + bx - 12$ has $x^2 - 4$ as a factor, where a and b are real constants. Which of the following is also a factor of $P(x)$?

a. $2x - 1$ ✓

b. $2x + 1$

c. $x + 3$

d. $2x - 3$

We can solve this problem using either polynomial structure inspection or the Factor Theorem. **Method 1: Polynomial Structure & Division** Since $x^2 - 4$ is a factor of $P(x)$, the quotient $Q(x)$ must be a quadratic polynomial such that: $P(x) = (x^2 - 4)Q(x)$ To match the leading term $2x^4$ and the constant term -12, $Q(x)$ must be of the form $2x^2 + kx + 3$, because: - Leading term: $x^2 \cdot 2x^2 = 2x^4$ - Constant term: $-4 \cdot 3 = -12$ Expanding $(x^2 - 4)(2x^2 + kx + 3)$, the x^3 term comes solel.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Sequences & Series

Consider the equation $a^{\sqrt{x}} \cdot b^{\sqrt{x}} = (ab)^{\sqrt{x}}$ and $a^{\sqrt{x}} \div b^{\sqrt{x}} = \left(\frac{a}{b}\right)^{\sqrt{x}}$ where $a, b > 0$ and $aeq1, beq1$. If $k = \sqrt{x}$, find the value of k that satisfies the equation $\left(\frac{3^{k^2+2k+1}}{2^{k^2+2k+1}}\right) \cdot \left(\frac{2^{2k+2}}{3^{2k+2}}\right) = \left(\frac{3}{2}\right)^{5k-1}$.

a. 3

b. 1/3

c. **1/5 ✓**

d. 5

Let the given equation be $\left(\frac{3^{k^2+2k+1}}{?^{k^2+2k+1}}\right) \cdot \left(\frac{2^{2k+2}}{3^{2k+2}}\right) = \left(\frac{3}{2}\right)^{5k-1}$. Using the property $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$, we can rewrite the left side of the equation as: $\left(\frac{3}{2}\right)^{k^2+2k+1} \cdot \left(\frac{2}{3}\right)^{2k+2} = \left(\frac{3}{2}\right)^{5k-1}$. Now, using the property $\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m$, we can rewrite.

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